

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12402571
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Gwon; Mi Jin

Drawer-type plant cultivation apparatus

Abstract

A drawer-type plant cultivation apparatus according to an embodiment of the present invention includes: a body unit configured such that a hexahedral frame is erected on the ground and a plurality of support bar pairs is symmetrically installed on inner surfaces of both sides of the hexahedral frame to be vertically spaced apart from each other; plant units each configured to provide at least one plant box that is provided such that both sides of the top end thereof are inserted into the body unit, so that the plant box is seated on a corresponding one of the support bar pairs; and piping units each disposed inside the plant box, and each comprising a piping part connected to the rear of the plant box to supply a nutrient solution and supply oxygen and one or more corrugated pipe parts connected to the bent points of the piping part.

Inventors:	Gwon; Mi Jin (Daegu, KR)
Applicant:	AG-UNI CO., LTD (Gyeongsan-si, KR)
Family ID:	86774553
Assignee:	AG-UNI CO., LTD (Gyeongsan-si, KR)
Appl. No.:	17/922316
Filed (or PCT Filed):	December 13, 2021
PCT No.:	PCT/KR2021/018903
PCT Pub. No.:	WO2023/113056
PCT Pub. Date:	June 22, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240315177 A1	Sep. 26, 2024

Foreign Application Priority Data

Publication Classification

Int. Cl.: A01G9/02 (20180101); A01G27/00 (20060101)

U.S. Cl.:

CPC A01G9/022 (20130101); A01G27/008 (20130101);

Field of Classification Search

CPC: A01G (31/06); A01G (9/023); A01G (27/006); A01G (27/008); A01G (9/24); A01G (31/02); A01G (9/02); A47B (2210/0056); A47B (88/40); A47B (88/417); A47B (2088/951); A47B (2088/421)

USPC: 211/85.23; 47/39; 47/83

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
1212471	12/1916	Forbes	312/334.8	A47B 88/427
2006/0191196	12/2005	Cochran	47/66.1	A01G 9/02
2015/0052808	12/2014	Wang	47/65.5	A01G 27/06
2015/0276098	12/2014	Garrett	156/195	B29C 48/131
2019/0335691	12/2018	Krakover	N/A	A01G 31/06
2020/0260668	12/2019	Choi	N/A	F25D 23/062
2022/0361420	12/2021	Lim	N/A	A01G 9/247
2023/0073964	12/2022	Hertel	N/A	A01G 9/023

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
104025992	12/2013	CN	N/A
100497941	12/2004	KR	N/A
100673967	12/2006	KR	N/A
101755812	12/2016	KR	N/A
1020180052229	12/2017	KR	N/A
102156744	12/2019	KR	A01G 31/02

OTHER PUBLICATIONS

International Search Report of International Patent Application No. PCT/KR2021/018903, mailed Aug. 26, 2022, 3 pages. cited by applicant

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is a national phase claiming the benefit of and priority to International Patent Application No. PCT/KR2021/018903, entitled DRAWER-TYPE PLANT CULTIVATION APPARATUS, filed Dec. 13, 2021, which claims priority to Korean Patent Application No. 10-2021-0177525, filed on Dec. 13, 2021, which are hereby incorporated by reference in their entirety.

TECHNICAL FIELD

(2) The present invention relates to a drawer-type plant cultivation apparatus that is capable of cultivating soil plants in a vertical structure.

BACKGROUND ART

(3) As living spaces and farmland are becoming narrower due to population growth and industrialization, efforts to cultivate a large quantity of flowers and vegetables in a small area are being carried out in various fields. Furthermore, as the residential form in large cities is converted from single and multi-family houses to apartments, there are a growing number of people who feel like being in the garden of a detached house and utilize plant cultivation for hobbies or decoration by cultivating plants such as flowering plants and flowers on the veranda of an apartment house or in the inside of a building.

(4) In general, to cultivate plants on the veranda of an apartment house or in the inside of a building, there is a method of cultivating plants in a cultivation box. Since a common cultivation box is intended to be installed on the ground, there is a problem in that a large cultivation space is unnecessarily occupied and thus space utilization is inefficient. Furthermore, when the space of the veranda of an apartment house or the inside of a building is narrow, the number of plants that can be cultivated is limited.

(5) In order to overcome these problems, a stacked flowerpot arrangement in which a plurality of flowerpots is vertically stacked one above another has been developed. However, this conventional stacked flowerpot arrangement has disadvantages in that the spaces between flowerpots cannot be utilized and it was inconvenient to take out plants planted in flowerpots and plant new plants.

(6) In particular, the conventional stacked flowerpot arrangement has shortcomings in that the structure thereof is complicated, the equipment cost thereof is high, and a deformed module or frame cannot be easily changed. Furthermore, the conventional stacked flowerpot arrangement has a structure in which flowerpots are stacked one above another, so that it is difficult to provide even lighting for each flowerpot and management such as watering is cumbersome, with the result that the conventional stacked flowerpot arrangement is not widely utilized.

(7) In order to overcome this problem, Korean Patent No. 10-1419119 provides a 'Vertical Greening Structure' in which basic planting members are slidably fitted into sliding grooves formed in posts, so that the installation of the basic planting members is considerably convenient and the replacement of plants is considerably easy.

(8) However, in the vertical greening structure, the sliding grooves are formed vertically along the posts, so that to separate the basic planting members for the replacement of plants, it is necessary to completely separate all basic planting members ranging from a stacked upper basic planting member to a corresponding basic planting member. Accordingly, this separation work is cumbersome and takes a lot of work time due to unnecessary work time.

(9) In addition, when replacing plants in the vertical greening structure, a manager has to lift and move the soil and basic planting members contained inside, so that it is difficult for the elderly or women to lift them up, thereby making it difficult to use the vertical greening structure, and so that there is a problem in that the lifting and moving work may lead to bodily injury to the manager.

(10) Meanwhile, generally, in the case of soil plants, the leaves of the plants absorb oxygen only during the nighttime, but the roots thereof can actively absorb fertilizer components only when they absorb oxygen during the daytime and nighttime. For example, the reason for removing paddy water before ear emergence during rice cultivation is to supply oxygen to the soil by exposing the bottom of a paddy field to the atmosphere, thereby promoting the absorption of potassium (K) and the like.

(11) However, the soil is always deprived of oxygen due to the respiration of roots and microorganisms and the oxidation of soil minerals, and also the soil is further deficient in oxygen because ventilation is blocked. Accordingly, roots become thin and plants are prone to disease, so that there are problems in that pesticides have to be continuously sprayed and, in severe cases, plants die.

PRIOR ART LITERATURE

Patent Document

(12) Korean Patent No. 10-1419119

DISCLOSURE

Technical Problem

(13) In order to overcome the above-described problems, an object of the present invention is to provide a drawer-type plant cultivation apparatus in which plant units planted with plants are inserted and installed in a body unit in a drawer-type structure, so that replacement work can be facilitated when plants or soil mixes are replaced, so that space utilization is excellent because the plant units can be selectively installed in the body unit according to the installation locations desired by a user or the growth height of plants, and so that no separate pipe separation work is required because the plant units can be moved forward and rearward without change in the original state in which piping units are installed.

Technical Solution

(14) According to an embodiment of the present invention, there is provided a drawer-type plant cultivation apparatus, including: a body unit configured such that a hexahedral frame is erected on the ground and a plurality of support bar pairs is symmetrically installed on the inner surfaces of both sides of the hexahedral frame to be vertically spaced apart from each other; plant units each configured to provide at least one plant box that is provided such that both sides of the top end thereof are inserted into the front of the body unit, so that the plant box is seated on a corresponding one of the support bar pairs on both sides and is movable forward and rearward; and piping units each disposed inside the plant box, and each including a piping part connected to the rear of the plant box to supply a nutrient solution and supply oxygen and one or more corrugated pipe parts connected to the bent points of the piping part; wherein, when the plant box is moved forward, the piping unit is moved forward of the body unit in an original state of being disposed in the plant box while the corrugated pipe parts are extended; and wherein, when the plant box is moved rearward, the piping unit is inserted into the body unit in the original state of being disposed in the plant box while the corrugated pipe parts are contracted.

(15) Furthermore, the body unit includes slide grooves formed of grooves having a set depth that are formed in the top surfaces of the support bar pairs in forward and rearward directions; the plant box includes slide protrusions formed of protrusions that are formed on the bottom surfaces of both sides of the top end thereof in a shape corresponding to that of the slide grooves of the body unit; and the plant box is selectively inserted onto any one of the plurality of support bar pairs according to the height of plants to be planted in the plant box.

(16) Furthermore, the body unit includes opening/closing doors that are installed on both sides of

the front of the hexahedral frame and selectively opened and closed to limit the slide movement of the plant box.

(17) Furthermore, the body unit includes: a support part formed of a flat plate, and installed on the bottom surface of the hexahedral frame; and rolling balls provided to be movable by being installed along the edge of the support part.

(18) Moreover, the plant box includes: a drawer case configured such that the top and front surfaces thereof are open and both sides of the top end thereof protrude outward and form seating bars; an inner case formed in the shape of a hexahedral box, and configured such that plants are planted in a soil contained therein, the inner case is inserted into the front of the drawer case, an air circulation space is provided in the lower portion thereof by being spaced apart from the bottom surface of the drawer case by a set height, and a plurality of ventilation holes is formed in the bottom surface thereof; and a separation plate installed on the bottom surface of the inside of the inner case, and made of a mesh network so that it prevents the soil from falling into the air circulation space in the inner case.

Advantageous Effects

(19) According to the drawer-type plant cultivation apparatus according to an embodiment of the present invention configured as described above, there are the following effects:

(20) The plant units planted with plants are inserted and installed into the body unit in the vertical direction in a drawer-type structure, that space utilization is so excellent; and upon replacement of plants or a soil mix, the plant unit slides forward without the need to be completely separated from the body unit or only the plant unit requiring work is thanks to the shape separated characteristics of the corrugated pipe parts of the piping unit, so that the work can be facilitated.

(21) Furthermore, the plurality of support bar pairs is vertically spaced apart from each other on both sides of the body unit, so that the plant units can be selectively installed on the support bar pairs according to the growth height of the plants placed therein, and so that the locations of the plant units can be easily changed, and thus the intervals at which the plant units are installed can be adjusted.

(22) Furthermore, the plant unit moves safely in the forward and rearward directions without shaking left and right by sliding in the state in which the slide grooves of the support bar pair, disposed on both sides of the body unit, and the slide protrusions of the plant unit engage with each other, so that stability can be increased; and the support bar pair of the body unit and the seating bars of the plant unit come into close contact with each other, so that supporting force can be improved.

(23) Furthermore, the opening/closing doors are installed on both sides of the front of the body unit and limit the forward and backward movement of the plant units, so that the plant units can be prevented from falling; and the support part of the body unit is equipped with the rolling balls, so that there is an advantage in that it is easy to move the body unit when the body unit is installed.

(24) Furthermore, the air circulation space is provided in the lower portion of each of the plant units and thus ventilation is performed, so that breathability can be improved; and the humidity of the soil can be controlled and also the oxygen demand required by the roots of plants can be optimally met by supplying oxygen to the soil, so that the growth status of plants can be improved.

(25) Furthermore, there is an effect in that a considerably wide range of crops grown based on a soil can be cultivated by the light emission modules. The reason for this is that the light saturation point required for growth differs for each crop and also the degree or location of lighting must vary according to the degree of growth.

(26) Moreover, there is an effect in that a cultivation prescription is customized according to a cultivation and growth environment for each crop through the light emission parts as well as the supply of a nutrient solution and the supply of oxygen via drainage and breathability.

Description

DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a perspective view of a drawer-type plant cultivation apparatus according to the present invention;
- (2) FIG. 2 is a perspective view showing the state in which the opening/closing doors of the body unit of FIG. 1 are opened;
- (3) FIG. 3 is a partially enlarged view of the drawer-type plant cultivation apparatus according to the present invention;
- (4) FIG. 4 is a view showing the configuration of a plant unit of the drawer-type plant cultivation apparatus according to the present invention;
- (5) FIG. 5 is a view showing the configuration of a piping unit of the drawer-type plant cultivation apparatus according to the present invention;
- (6) FIG. 6 is a rear view of the drawer-type plant cultivation apparatus according to the present invention; and
- (7) FIG. 7 is a view showing the state in which the opening/closing doors of the piping compartment parts of the drawer-type plant cultivation apparatus according to the present invention are opened.

MODE FOR INVENTION

- (8) Embodiments of the present invention will be described in detail below with reference to the accompanying drawings so that those of ordinary skill in the art to which the present invention pertains can easily practice them. The present invention may be implemented in many different forms and is not limited to the structures or methods described herein.
- (9) An embodiment of the present invention relates to a drawer-type plant cultivation apparatus in which plant units are inserted and installed in a body unit in the form of drawers, so that replacement work can be facilitated by sliding the plant units in the forward and rearward directions when plants or soil mixes are replaced. Hereinafter, the drawer-type plant cultivation apparatus will be referred to as the 'plant cultivation apparatus.'
- (10) FIG. 1 is a perspective view of a drawer-type plant cultivation apparatus according to the present invention, and FIG. 2 is a perspective view showing the state in which the opening/closing doors of the body unit of FIG. 1 are opened.
- (11) Referring to FIGS. 1 and 2, the plant cultivation apparatus according to the embodiment of the present invention is an apparatus in which plants are arranged in a vertical structure, so that space utilization is increased and the growth environment of plants is improved. The plant cultivation apparatus may include: a body unit **100** configured to provide a vertical space in which plants are mounted; plant units **200** configured to allow a plurality of plant boxes **201** planted with plants to be mounted in the body unit; piping units **300** disposed in the respective plant units, and configured to supply and discharge a nutrient solution and supply oxygen; and piping compartment parts **400** each configured to provide a space through which pipes of a piping installed in the plant unit are inserted into the rear portion of the body unit. Hereinafter, the nutrient solution supplied to the plant units **200** will be described and explained as the 'nutrient solution.' The nutrient solution includes all types of liquid supplied to plants, but is not limited thereto.
- (12) The body unit **100** is a means for providing a space in which soil plants are disposed. In detail, the body unit **100** may include: a hexahedral frame **110** formed in a vertically erected form on the ground, and configured such that a space is formed therein; and a plurality of support bar pairs **120** installed to be spaced apart in the vertical direction so that the plant units **200** or light emission parts **140** can be mounted in the space of the hexahedral frame **110**. In this case, the support bar pairs **120** are each provided such that a pair of square bars are symmetrically installed on both sides of the hexahedral frame **110** so that both ends of one of the plant units **200** can be seated thereon,

and slide grooves **121** having a set depth may be formed in the top surfaces of the support bar pairs **120** in the forward and rearward longitudinal directions.

(13) In other words, the body unit **100** is provided such that each of the plant units **200** can be installed on any one of the plurality of support bar pairs **120**, so that a location can be changed by taking into consideration the installation location of the plant unit **200** according to the growth height of the plants of the plant unit **200**. Furthermore, the body unit **100** may include pairs of holders **130** that are inserted into the slide grooves **121** of the support bar pairs **120**. Each holder of the pairs of holders **130** is configured such that one end thereof is formed of a flat plate having the thickness of the slide groove **121**, bent in an inverted 'L' shape and inserted and a cylindrical insertion hole is formed and provided at the other end thereof. In this case, each pair of holders **130** selectively slide forward and rearward along the slide grooves **121** of the support bar pair **120**, receive both ends of the light emission part **140** through the insertion holes to facilitate installation, and may be easily detached and installed when the installation location of the plant unit **200** is changed.

(14) The light emission parts **140** of the body unit **100** are configured to provide light so that lighting is provided to the upper parts of the plant units **200**. In this case, the light emission parts **140** may be installed above the plant boxes **201** in order to overcome the problem in which uniform light is not provided to the plant boxes **201** due to the vertical structure of the plant cultivation apparatus. In particular, the light emission parts **140** of the body unit **100** have a structure that can be mounted using the support bar pairs **120** on which the plant boxes **201** are mounted, so that even when each of the plant boxes **201** is installed anywhere in the body unit **100**, the light emission part **140** can be installed on the support bar pair **120** above the location at which the plant box **201** is installed, thereby increasing the utilization of the support bar pairs **120**.

(15) Furthermore, the body unit **100** may include: opening/closing doors **150** rotatably installed on both sides of the front of the hexahedral frame **110**, and configured to be selectively opened and closed; a support part **160** formed in the shape of a flat plate, and configured to support the bottom surface of the hexahedral frame **110**; wheel-shaped rolling balls **170** installed along the edge of the bottom surface of the support part **160**; and ventilation holes **180** formed on both sides of the hexahedral frame **110** and configured to allow air to be circulated through the inside of the hexahedral frame **110**. In particular, the opening/closing doors **150** are installed on both sides of the front of the body unit **100** and limit the forward and rearward movement of the plant units **200**, so that there is an advantage of preventing the plant units **200** from falling, and the support part **160** of the body unit **100** is equipped with the rolling balls **170**, so that there is an advantage in that it is easy to move the plant cultivation apparatus when the plant cultivation apparatus is installed.

(16) FIG. 3 is a partially enlarged view of the drawer-type plant cultivation apparatus according to the present invention, and FIG. 4 is a view showing the configuration of a plant unit of the drawer-type plant cultivation apparatus according to the present invention.

(17) Referring to FIGS. 3 and 4, the plant unit **200** is a means for providing a space through which plants can be planted and grown in at least one plant box **201**. The plant units **200** are vertically spaced apart from each other in the inside of the body unit **100**, and the plant box **201** is selectively inserted onto any one of the plurality of support bar pairs **120** according to the growth height of plants to be planted in the plant box **201**, so that the interval between the plant boxes **201** is adjusted. In other words, the locations at which the plant units **200** are installed in the body unit **100** may be freely changed, and there is no restriction on the installation space, so that one or more plant boxes **201** may be installed in one tier, in two tiers, in three tiers, or the like by a user.

(18) In particular, the plant units **200** are inserted and installed into the body unit **100** in a drawer-type structure, which is excellent in terms of space utilization. When plants or a soil mix are replaced, work may be facilitated by sliding the plant unit **200** forward in the state of being seated on the support bar pair **120** without the need to be completely separated from the body unit **100** or by separating only the plant box **201** requiring the work. In detail, each of the plant boxes **201** may

include a drawer case **210** configured to be seated on the support bar pair **120** of the body unit **100**, an inner case **220** configured such that a soil is contained and plants are planted therein, and a separation plate **230** configured to be installed on the bottom of the inner side of the inner case **220**. (19) The drawer case **210** is shaped in the form of a square box with an open top and front, and is provided with seating bars **211** with both upper ends protruding outward. In this case, the drawer case **210** may have a slide surface formed on the bottom surface thereof or include slide protrusions (not shown) protruding in a shape corresponding to that of the slide grooves **121** of the body unit **100** so that the seating bars **211** can come into close contact with and move smoothly along the support bar pair **120**. In this drawer case **210**, the plant unit **200** is allowed to move in the forward and rearward directions without shaking left and right while the slide grooves **121** of the support bar pairs **120** and the slide protrusions of the plant unit **200** engage with each other and slide, thereby increasing stability.

(20) In addition, the drawer case **210** is equipped with a plurality of first through holes **213** on the rear side thereof so that the piping unit **300** can be fixed therethrough, so that the piping unit **300** is inserted through the first through holes **213** and provided to be coupled to and separated from the outside. Furthermore, the drawer case **210** has guide bars **212** protruding from both sides of the open front in order to have a set inward width, so that corresponding sides of the drawer case **210** and the inner case **220** are formed to be spaced apart from each other by the set width. For this reason, the guide bars **212** of the drawer case **210** may guide the inner case **220** without shaking left and right by guiding the inner case **220** from both sides when the inner case **220** is inserted into the drawer case **210**.

(21) The inner case **220** is shaped in the form of a square box with an open top, and is inserted into the front of the drawer case **210** with plants planted in a soil. In this case, the inner case **220** is installed to be spaced apart from the bottom surface of the drawer case **210** by a set height, and an air circulation space **222** is formed between the drawer case **210** and the inner case **220**. Furthermore, a plurality of ventilation holes **223** having a diameter equal to or larger than a set size is formed through the bottom surface of the inner case **220**, so that the soil can be ventilated into the air circulation space **222** through the ventilation holes **223**.

(22) The inner case **220** is provided with a plurality of seating cutouts **224** formed at the top ends of both side and rear surfaces thereof, and a plurality of second through holes **225** at locations corresponding to those of the first through holes **213** of the drawer case **210**. Furthermore, the inner case **220** allows the piping unit **300** to pass through the first through holes **213** of the drawer case **210** in the state of being seated in the seating cutouts **224** or being inserted into the second through holes **225**, and thus allows the drawer case **210** and the inner case **220** to be moved in an integrated manner or to be separated and moved. In addition, the inner case **220** may include a front cover **221** that is disposed on the front thereof, formed to protrude forward, and formed so that hands can be inserted into both sides and both sides serve as grips.

(23) The front cover **221** may include seating facets **221a** formed to protrude to both sides and space facets **221b** formed to protrude downward. The seating facets **221a** are provided to come into close contact with the seating bars **211** of the drawer case **210** and to be installed in a stable posture. Furthermore, the space facets **221b** are provided so that the air circulation space **222**, which is a space spaced apart from the bottom of the inner case **220**, can be formed, and are provided to be supported on the bottom surface of the inside of the drawer case **210**. In addition, a crop planting distance scale is marked on the top surface of the front cover **221**, so that a manager can directly plant plants at optimal distances while adjusting the intervals between the plants to be planted in the soil.

(24) The separation plate **230** is installed on the bottom surface of the inside of the inner case **220**, and may be formed of a mesh in which mesh holes having a size smaller than that of the particles of the soil are formed. The separation plate **230** may prevent the soil of the inner case **220** from falling into the air circulation space **222** through the ventilation holes **223** formed under the

separation plate **230**. In addition, the separation plate **230** is provided to be supported from the bottom surface of the inner case **220** in which the ventilation holes **223** are formed and thus withstand the weight of the soil. In particular, the separation plate **230** may prevent part of the soil from being mixed with a nutrient solution as the nutrient solution is collected into the air circulation space **222** after it has been supplied to the soil, so that the collected nutrient solution can be reused and the clogging of a pipe can be prevented from occurring.

(25) FIG. 5 is a view showing the configuration of a piping unit of the drawer-type plant cultivation apparatus according to the present invention.

(26) Referring to FIG. 5, the piping unit **300** is installed inside the plant box **201**, and is a means for supplying or discharging a nutrient solution to or from the soil and supplying oxygen. In detail, the piping unit **300** is disposed inside the plant box **201**, and may include a piping part **301** connected to the rear of the plant box **201** to supply a nutrient solution and supply oxygen, and corrugated pipe parts **302** formed in the shape of bellows and connected to the bent points of the piping part **301**. The piping unit **300** may be moved forward in the state in which the piping unit **300** is installed in the plant box **201** thanks to the shapes of the corrugated pipe parts **302** without change. A detailed operation sequence of this will be described below.

(27) First, in the piping unit **300**, when the plant box **201** is pulled forward upon replacement of plants or a soil (a cultivation soil) by a manager, the piping unit **300** is moved together forward of the body unit **100** in the state of being disposed in the plant box **201** as the corrugated pipe parts **302** are extended, so that the replacement of plants or a soil can be performed without separate pipe separation work. Furthermore, in the piping unit **300**, when the plant box **201** is pushed backward and moved by a manager, the piping unit **300** slides, and is inserted and installed into the body unit **100** in the state of being disposed in the plant box **201** as the corrugated pipe parts **302** are contracted.

(28) The piping unit **300** may include a nutrient solution module **310** configured to supply a nutrient solution to a soil, an oxygen module **320** configured to supply oxygen, and a drainage module configured to discharge a nutrient solution to the outside. The nutrient solution module **310**, the oxygen module **320**, and the drainage module of the piping unit **300** are extended to the rear end of the plant unit **200** and are respectively connected to single pipes. Bent points are formed of the bellows-shaped corrugated pipe parts **302**, so that the plant box **201** can be smoothly moved in the state in which the piping unit **300** is connected. In this case, the piping unit **300** is configured in a modular form in which the nutrient solution module **310**, the oxygen module **320**, and the drainage modules **330** may be assembled to correspond to the number of plant boxes **201**. The piping unit **300** is disposed to be coupled to or detached from the outside of the rear end of the plant unit **200**.

(29) The nutrient solution module **310** is a piping module in which external irrigation pipes **311** disposed in a plurality of rows on the surface of a soil and underground irrigation pipes **312** disposed in a plurality of rows in the lower portion of the inside of the soil are connected to each other and simultaneously supply a nutrient solution to the surface and inside of the soil. In this case, in the nutrient solution module **310**, a nutrient solution supply pipe **313** having a single flow path is connected to the rear portion of the external irrigation pipes **311** so that a nutrient solution is supplied from the outside, uniformly distributed into the external irrigation pipes **311** and the underground irrigation pipes **312**, and then supplied to the soil. Although such a nutrient solution module **310** has not been able to maintain the water retention capacity by allowing a nutrient solution to be supplied only to the surface of a soil in the prior art, the nutrient solution module **310** is configured in two tiers so that a nutrient solution is uniformly supplied to the inside and outside of the soil simultaneously, thereby increasing the water retention capacity and also improving the growth rate of plants.

(30) The oxygen module **320** is a piping module in which pipes are arranged in a plurality of columns in the bottom portion of the soil under the underground irrigation pipes **312** and supply

oxygen to the soil. Furthermore, in the oxygen module **320**, single oxygen pipes **321** arranged in a plurality of columns under the underground irrigation pipes **312** and extending to the outside through the first and second through holes **213** and **225** of the plant box **201** and an oxygen supply pipe **322** having a single flow path are connected to each other through the rear portion of the plant box **201**, are supplied with oxygen from the outside, and uniformly supply the oxygen to the overall area of the soil. The oxygen module **320** controls the humidity of the soil when the soil is excessively humid by spraying oxygen into the soil, and allows the standard oxygen demand required by the growth of plants to be met. In particular, the oxygen module **320** induces air circulation in the soil through continuous oxygen supply, and helps the air to be moved to the lower air circulation space **222** and to be circulated again.

(31) The drainage module is disposed in the air circulation space **222** of the plant box **201**, and is a module configured to collect and discharge a nutrient solution to the outside. In this case, the drainage module may be inserted into the air circulation space **222** of the plant box **201** in the shape of a drip tray. In this case, via the drainage module, a nutrient solution may be collected as it falls into the drip tray through the ventilation hole **223**, and a manager may separate the inner case **220** from the drawer case **210** and discharge the nutrient solution, collected in the drip tray, to the outside. Furthermore, the drainage module may be formed in such a manner that a nutrient solution discharge pipe is extended rearward into the air circulation space **222** of the plant box **201**, and may allow a nutrient solution to be discharged to the outside through the nutrient solution discharge pipe. As described above, the drainage module may be configured in the same shape and structure as the oxygen module **320**, and does not limit the structure through which a nutrient solution is discharged.

(32) FIG. **6** is a rear view of the drawer-type plant cultivation apparatus according to the present invention, and FIG. **7** is a view showing the state in which the opening/closing doors of the piping compartment parts of the drawer-type plant cultivation apparatus according to the present invention are opened.

(33) Referring to FIGS. **6** and **7**, the piping compartment parts **400** are means for arranging the pipes of the piping units **300** or the electric wires of the light emission parts **140** in order to prevent them from being exposed to the outside. In detail, the piping compartment parts **400** are formed on both sides of the rear of the hexahedral frame **110**, and provide storage spaces **401** into which the nutrient solution modules **310**, the oxygen modules **320**, and the drainage modules installed in the plant boxes **201** are inserted not to be exposed to the outside. In one of the piping compartment parts **400**, nutrient solution supply pipes **313** installed in one or more plant boxes may be detachably connected to a single flow path and installed in a modular form in the storage space **401**, and oxygen supply pipes **322** or nutrient solution discharge pipes may also be installed in the storage space **401** in the same module form.

(34) The two piping compartment parts **400** may include a plurality of insertion guide holes **410** in which holes larger than the diameters of the nutrient solution module **310**, the oxygen module **320** and the drainage module are formed through the opposite surfaces of piping compartment parts **400** in the vertical directions. In this case, the plurality of insertion guide holes **410** are provided such that the piping units **300** can be inserted into the insertion guide holes **410** at optimal locations according to the installation locations of the plant boxes **201**. Furthermore, the piping compartment parts **400** may further include respective opening/closing doors **420** with handles **421** that are rotatably installed on the rear sides and allow the nutrient solution modules **310**, the oxygen modules **320**, and the drainage modules to be selectively connected and disconnected by a manager according to the installation locations of the plant boxes **201**.

(35) The piping compartment parts **400** may be opened by a manager in order to change the installation locations of a piping when the locations of one or more plant boxes **201** are changed, and the lengths of the connection pipes of one or more plant units **200** disposed in the piping compartment parts **400** may be easily changed according to the intervals between the plant boxes

201. The piping compartment parts 400 may allow pipes and wires to be stored therein while minimizing the exposure of the pipes and the wires to the outside, thereby improving the aesthetics of the plant cultivation apparatus and also allowing only a manager to access the pipes and the wires, so that the apparatus can be used safely.

(36) The following implementations were made to test changes in the moisture content of a soil when a nutrient solution was supplied to the surface and inside of the soil simultaneously by the nutrient solution module of the plant cultivation apparatus of the present invention and oxygen was supplied to the inside of the soil by using the oxygen module, as follows. However, the following examples are only illustrative of the present invention, and the spirit of the present invention is not limited to the following examples.

Example 1

(37) In order to check changes in the moisture content of soil the during plant cultivation using the plant cultivation apparatus of the present invention, a soil with a bulk density of less than 0.3 mg/m³, a pH of 5 to 7, an EC of 1.2 dS/m and 100 L was placed in a plant box with a size of 1000×500×500 mm³ and changes in the soil were measured at 10-minute intervals while water and oxygen were passed through the soil by using a feed pump with a pressure of 8.3 bar and a flow rate of 10 to 12 LPM and a compressor with a horsepower of 0.8 and a capacity of 9 L. The CM2105 model was used as a sensor for measuring the temperature and humidity of the outside, and the KSM-8900 model was used as a sensor for measuring the inside of the soil.

(38) For the experiments intended to check changes in the moisture content of the soil, Example 1 in which water was simultaneously supplied to the surface and inside of the soil and oxygen was supplied to the inside of the soil according to the present invention and Comparative Example 1 in which water was supplied only to the surface of the soil were distinguished from each other.

(39) TABLE-US-00001 TABLE 1 Comparative Example 1 External External Environment Environment Soil Temperature Humidity Humidity 1st Measurement 24° C. 40.2% 34% 2nd Measurement 23.9° C. 40.4% 32.2%.sup. 3rd Measurement 24° C. 40.2% 30.3%.sup. 4th Measurement 24° C. 40.5% 29.8%.sup. 5th Measurement 24.1° C. 41.5% 29% 6th Measurement 24.3° C. 40.6% 29% 7th Measurement 24.4° C. 40.6% 29% 8th Measurement 24.5° C. .sup. 40% 28.1%.sup. 9th Measurement 24.4° C. .sup. 41% 28% 10th Measurement 24.6° C. 40.8% 28%

(40) TABLE-US-00002 TABLE 2 Example 1 External External Environment Environment Soil Temperature Humidity Humidity 1st Measurement 24.7° C. 44.6% 34% 2nd Measurement 24.6° C. 43.9% 33% 3rd Measurement 24.6° C. 46.2% 32.1%.sup. 4th Measurement 24.6° C. 46.9% 31.5%.sup. 5th Measurement 25.4° C. 40.9% 31% 6th Measurement 26.2° C. 39.2% 31% 7th Measurement 26.6° C. 38.4% 30.3%.sup. 8th Measurement 26.° C. 38.1% 30% 9th Measurement 26.5° C. 37.8% 30% 10th Measurement 26.5° C. 37.9% 30%

(41) Regarding the comparison between Comparative Example 1 of Table 1 and Example 1 of Table 2, when measurements were made after water was supplied to the soil every 10 minutes, the soil humidity of Example 1 was measured as being preserved by about 2% more than that of Comparative Example 1. Accordingly, it can be found that the water retention capacity of the soil was improved by simultaneously spraying water to the surface and inside of the soil and supplying oxygen according to the present invention.

Example 2

(42) In order to check the promotion of plant growth during plant cultivation using the plant cultivation apparatus of the present invention, a soil with a bulk density of less than 0.2 mg/m³, a pH of 6 to 7, an EC of 0.8 dS/m and 100 L was placed in a plant box with a size of 1000×500×500 mm³ and the changes of chamomile were measured at 5-day intervals while water and oxygen were passed through the soil by using a feed pump with a pressure of 8.3 bar and a flow rate of 10 to 12 LPM and a compressor with a horsepower of 2.5 and a capacity of 24 L.

(43) For the experiments intended to check the promotion of the growth of chamomile, Example 2 in which water was simultaneously supplied to the surface and inside of the soil and oxygen was

supplied to the inside of the soil according to the present invention and Comparative Example 2 in which water was supplied only to the surface of the soil were distinguished from each other.

(44) TABLE-US-00003 TABLE 3 Comparative Example 2 Number Plant Length Leaf Length Leaf Width of Leaves 1st Measurement 8 mm 8.6 mm 1.2 mm 6 leaves 2nd Measurement 11 mm 10 mm 2.1 mm 6 leaves 3rd Measurement 14.5 mm 15.8 mm 2.6 mm 9 leaves

(45) TABLE-US-00004 TABLE 4 Example 2 Number Plant Length Leaf Length Leaf Width of Leaves 1st Measurement 9 mm 9.5 mm 1.5 mm 6 leaves 2nd Measurement 12 mm 13 mm 2.5 mm 7 leaves 3rd Measurement 15.5 mm 17.5 mm 3 mm 10 leaves

(46) Regarding the comparison between Comparative Example 2 of Table 3 and Example 2 of Table 4, when measurements were made at 5-day intervals after the soil had been supplied with water, the plant length, leaf length, and leaf width of Example 2 were measured as having grown about 0.4 to 1 mm larger than those of Comparative Example 2, and the number of leaves was measured as having grown by one more in the 2nd and 3rd measurements. Accordingly, it can be found that the growth rate of plants was increased by simultaneously spraying water to the surface and inside of the soil and supplying oxygen according to the present invention.

(47) According to the plant cultivation apparatus according to the embodiment of the present invention, the following effects are obtained:

(48) The plant units **200** planted with plants are inserted and installed into the body unit **100** in the vertical structure, so that space direction in a drawer-type utilization is excellent; and upon replacement of plants or a soil mix, the plant unit **200** slides forward without the need to be completely separated from the body unit **100** or only the plant unit **200** requiring work is separated thanks to the shape characteristics of the corrugated pipe parts **302** of the piping unit **300**, so that the work can be facilitated.

(49) Furthermore, the plurality of support bar pairs **120** is vertically spaced apart from each other on both sides of the body unit **100**, so that the plant units **200** can be selectively installed on the support bar pairs **120** according to the growth height of the plants placed therein, and so that the locations of the plant units **200** can be easily changed, and thus the intervals at which the plant units **200** are installed can be adjusted.

(50) Furthermore, the plant unit **200** moves safely in the forward and rearward directions without shaking left and right by sliding in the state in which the slide grooves **121** of the support bar pair **120**, disposed on both sides of the body unit **100**, and the slide protrusions of the plant unit **200** engage with each other, so that stability can be increased; and the support bar pair **120** of the body unit **100** and the seating bars **211** of the plant unit **200** come into close contact with each other, so that supporting force can be improved.

(51) Furthermore, the opening/closing doors **150** are installed on both sides of the front of the body unit **100** and limit the forward and backward movement of the plant units **200**, so that the plant units **200** can be prevented from falling; and the support part **160** of the body unit **100** is equipped with the rolling balls **170**, so that there is an advantage in that it is easy to move the body unit **100** when the body unit **100** is installed.

(52) Furthermore, the air circulation space **222** is provided in the lower portion of each of the plant units **200** and thus ventilation is performed, so that breathability can be improved; and the humidity of the soil can be controlled and also the oxygen demand required by the roots of plants can be optimally met by supplying oxygen to the soil, so that the growth status of plants can be improved.

(53) Furthermore, the water retention capacity of the soil is improved by simultaneously supplying a nutrient solution to the soil surface and inside of the plant unit **200** by the nutrient solution module **310**, so that the growth rate can be improved; and the air circulation space **222** is formed under the soil and thus internal air is changed, so that the growth environment of plants can be improved.

(54) Furthermore, oxygen is supplied to the lower portion of a soil by each oxygen modules **320**, so that the humidity of the soil can be controlled, so that the oxygen required for the growth of plants

is directly supplied in such a manner that the oxygen is directly absorbed through the roots of the plants, and so that the oxygen demand required by the roots of plants can be optimally met and thus oxygen can be smoothly supplied to the roots.

(55) Furthermore, each of the nutrient solution module **310**, the oxygen module **320**, and the drainage module is detachably installed in each plant box **201** in a modular form, so that the location of the plant box **201** installed in the body unit **100** can be changed freely, and so that the intervals at which the plant boxes **201** are installed can be adjusted according to the growth height of plants.

(56) Furthermore, the light emission part **140** is provided above each of the plant boxes **201** so that lighting can be provided to each of the plant boxes **201** arranged in a vertical structure, so that even lighting is achieved and thus plants can grow uniformly and normally, and so that the location of the light emission part **140** can be freely changed according to the location at which the plant box **201** is installed.

(57) Moreover, the nutrient solution modules **310**, the oxygen modules **320**, and the drainage modules installed in the plant cultivation apparatus are inserted into the piping compartment parts **400** along optimal paths according to the locations at which the plant boxes **201** are installed, so that aesthetics can be improved by storing pipes inside the piping compartment parts **400** while minimizing exposure to the outside, and so that the pipes can be stored safely by allowing only a manager to access it.

(58) The present invention has been described so far with a focus on the preferred embodiments. It will be understood by those of ordinary skill in the art to which the present invention pertains that the present invention may be implemented in a modified form without departing from the essential characteristics of the present invention. Therefore, the disclosed embodiments should be taken into consideration in an illustrative sense rather than a restrictive sense. The scope of the present invention is defined based on the attached claims rather than the foregoing detailed description, and all differences falling within the scopes equivalent to the claims should be construed as being included in the present invention.

Claims

1. A drawer-type plant cultivation apparatus, comprising: a body unit having a hexahedral frame configured to be erected on a ground, the body unit having a plurality of support bar pairs symmetrically disposed on inner surfaces of both sides of the hexahedral frame to be vertically spaced apart from each other; a plant unit having at least one plant box, each of the at least one plant box having both sides of a top end thereof being configured to be inserted into a front of the body unit to be seated on one of the plurality of support bar pairs on both sides, the at least one plant box being movable forward and rearward; and at least one piping unit, each being disposed inside each of the at least one plant box, wherein each of the at least one piping unit comprises: a piping part connected to a rear of each of the at least one plant box to supply a nutrient solution and supply oxygen; and one or more corrugated pipe parts connected to bent points of the piping part, wherein, when the at least one plant box is moved forward, the at least one piping unit is configured to be moved forward of the body unit while being disposed in the at least one plant box while the one or more corrugated pipe parts are extended, wherein, when the at least one plant box is moved rearward, the at least one piping unit is configured to be inserted into the body unit while being disposed in the at least one plant box while the one or more corrugated pipe parts are contracted, wherein each of the at least one plant box is configured to be selectively installed onto any one of the plurality of support bar pairs according to a height of a plant to be planted in each of the at least one plant box, the at least one plant box having no restriction to installation space from a bottom of the body unit to a top of the body unit, wherein each of the at least one plant box includes a plurality of through holes defined on a bottom portion of a rear side of each of the at

least one plant box, and each of the at least one piping unit includes an oxygen module including a plurality of oxygen pipes, wherein the plurality of oxygen pipes extends through the plurality of through holes, respectively, to be arranged in a bottom portion of a soil contained in each of the at least one plant box to supply oxygen uniformly to an overall area of the soil.

2. The drawer-type plant cultivation apparatus of claim 1, wherein the body unit comprises doors disposed on both sides of a front of the hexahedral frame, the doors being configured to be selectively opened and closed to limit a slide movement of the at least one plant box.

3. The drawer-type plant cultivation apparatus of claim 1, wherein the body unit comprises: a support part having a flat plate and disposed on a bottom surface of the hexahedral frame; and rolling balls that are movable and disposed along an edge of the support part.

4. The drawer-type plant cultivation apparatus of claim 1, wherein each of the at least one plant box comprises: a drawer case having top and front surfaces thereof that are open, the drawer case having both sides of a top end thereof protruding outward and forming seating bars; an inner case having a shape of a hexahedral box, the inner case being configured to have plants planted in a soil therein, wherein the inner case is configured to be inserted into a front of the drawer case, an air circulation space is defined in a lower portion of the inner case by being spaced apart from a bottom surface of the drawer case by a set height, and a plurality of ventilation holes is defined in a bottom surface of the inner case; and a separation plate disposed on a bottom surface of an inside of the inner case, the separation plate having a mesh network to prevent the soil from falling into the air circulation space in the inner case.
